Conics: Composite Transformations

Questions 1–25

Question 1. Write down the equation of the image of the ellipse given by

$$\frac{x^2}{3} + \frac{y^2}{4} = 1$$

when it has been transformed by each of the following transformations, giving your answer in the form

$$\frac{(x-p)^2}{a} + \frac{(y-q)^2}{b} = 1,$$

where a, b, p and q are constants to be found.

a) Rotated by $\frac{\pi}{2}$ radians anti-clockwise about $(0,0)$ followed by an enlargement of scale factor 2 with centre $(0,0)$.

b) A reflection in the x-axis, followed by a rotation by $\frac{\pi}{2}$ radians anti-clockwise about $(0,0)$.

c) A translation by vector $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$, followed by a reflection in the line $y = -x$.

d) Rotated by $\frac{3\pi}{2}$ radians anti-clockwise about $(0,0)$ followed by an enlargement of scale factor $\frac{1}{2}$ with centre $(0,0)$.

e) Stretch parallel to the x-axis, scale factor 3, followed by a translation by vector $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$, followed by a rotation of $\frac{3\pi}{2}$ radians anti-clockwise about $(0,0)$.

Question 2. Write down the equation of the image of the hyperbola given by

$$\frac{x^2}{6} - \frac{y^2}{3} = 1$$

when it has been transformed by each of the following transformations:

a) Rotated by $\frac{3\pi}{2}$ radians anti-clockwise about $(0,0)$ followed by a stretch parallel to the y-axis, scale factor 4.

b) A reflection in the y-axis, followed by a rotation by $\frac{\pi}{2}$ radians anti-clockwise about $(0,0)$.

c) A translation by vector $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$, followed by a reflection in the line $y = x$.

d) Stretch parallel to the x-axis, scale factor $\frac{1}{3}$, followed by a reflection in $y = -x$.

e) Enlargement of scale factor 2 with centre $(0,0)$, followed by a translation by vector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, followed by a rotation of $\frac{\pi}{2}$ radians anti-clockwise about $(0,0)$.

Question 3. Write down the equation of the image of the rectangular hyperbola given by $xy = 16$ when it has been transformed by each of the following transformations:

a) Rotated by $\frac{3\pi}{2}$ radians anti-clockwise about $(0, 0)$ followed by an enlargement of scale factor 3 with centre $(0, 0)$.

b) A reflection in the x-axis, followed by a rotation by $\frac{\pi}{2}$ radians anti-clockwise about $(0, 0)$.

c) A translation by vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, followed by a reflection in the line $y = -x$.

d) Stretch parallel to the x-axis, scale factor 2, followed by a translation by vector $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$.

e) Enlargement of scale factor 2 with centre $(0, 0)$, followed by a translation by vector $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$, followed by a rotation of $\frac{3\pi}{2}$ radians anti-clockwise about $(0, 0)$.

Question 4. Each of the following curves are transformed by a stretch parallel to the y-axis scale factor 2, followed by a translation by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$, followed by a rotation of π radians anti-clockwise about $(0, 0)$. Find the equation of the image:

a) $\frac{x^2}{5} + \frac{y^2}{12} = 1$

b) $\frac{x^2}{4} - \frac{y^2}{6} = 1$

c) $xy = 11$

d) $x^2 - y^2 = 9$

Question 5. Describe a pair of transformations that map:

a) $x^2 + 9y^2 = 1$ to $\frac{(x-2)^2}{36} + \frac{y^2}{4} = 1$
b) $\frac{x^2}{9} - \frac{y^2}{4} = 1$ to $\frac{(x-3)^2}{9} - \frac{y^2}{16} = 1$
c) $xy = 1$ to $(x-2)y = 3$
d) $\frac{x^2}{16} + \frac{y^2}{9} = 1$ to $\frac{(x-1)^2}{16} + \frac{(y+3)^2}{36} = 1$
e) $\frac{x^2}{4} - \frac{y^2}{5} = 1$ to $\frac{(2-x)^2}{8} - \frac{y^2}{18} = 1$
f) $\frac{x^2}{25} + \frac{y^2}{4} = 1$ to $\frac{(x+2)^2}{4} + \frac{(y-1)^2}{25} = 1$
g) $xy = 1$ to $(x+1)(y-2) = 4$
h) $\frac{y^2}{16} - \frac{x^2}{9} = 1$ to $9(y-1)^2 - 4x^2 = 144$
i) $\frac{x^2}{5} + \frac{y^2}{3} = 1$ to $x^2 + 5(y^2 - 4y) = -5$
j) $x^2 - \frac{y^2}{5} = 1$ to $x^2 = 5(y^2 + 2x)$

Question 6. The curve C_1, given by equation $\frac{x^2}{2} + \frac{y^2}{4} = 1$, is mapped to curve C_2 with equation $2(x-1)^2 + (y+3)^2 = 16$.

a) Describe the transformations that map curve C_1 to curve C_2.
b) Find where the curve C_2 intersects with the coordinate axes.
c) Hence, sketch the curve C_2.

Question 7. A curve C has equation $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

a) State the type of curve C is.

b) State the asymptotes of C.

c) Given that C is translated by vector $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$ followed by a reflection in the line $y = -x$, find the asymptotes of the image of C.

Question 8. The curve $x^2 + 4y^2 = 1$ is transformed by the matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ followed by $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

a) Describe the geometrical transformations represented by matrix $M = AB$.

b) Find the equation of the image of the ellipse under the transformation given by matrix M.

Question 9. The curve $x^2 - y^2 = 4$ is transformed by the matrix $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ followed by $B = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$.

a) Describe the geometrical transformations represented by matrix $M = AB$.

b) Find the equation of the image of the curve under the transformation given by matrix M.

Question 10. A curve C has equation $\frac{x^2}{9} - \frac{y^2}{4} = 1$. The curve C is transformed onto curve C' by the transformation represented by matrix A. In each of the following cases, find the equations of the asymptotes of curve C':

a) $A = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$

b) $A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$

c) $A = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 & -1 \\ 3 & 1 \end{pmatrix}$

d) $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$

Question 11. The ellipse E given by equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is transformed to an ellipse E' by the matrix $M = \frac{1}{3}\begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$.

a) Find the determinant of M.

b) Find the equation of the ellipse E'.

c) It is known that the area enclosed by the ellipse E is 6π. Find the area enclosed by the curve E'.

Question 12. The curve C given by equation $x^2 + y^2 = 9$ is transformed to curve C' by the matrix $M = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}$.

a) Find the determinant of M.

b) Show that the equation of the curve C' is given by $ax^2 - bxy + cy^2 = 81$, where a, b and c are constants to be found.

c) Find the area enclosed by the curve C'.

Question 13. The curve C given by equation $4x^2 - 8xy + 8y^2 = 1$ is transformed to curve C' by the matrix $M = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$.

a) Find the equation of the curve C'.
b) Find the area enclosed by the curve C'.
c) Hence, find the area enclosed by the curve C.

Question 14. The curve C_1, given by equation $\frac{x^2}{16} - \frac{y^2}{4} = 1$, is mapped to curve C_2 with equation $x^2 + 8x - y^2 = 0$.

a) State the equations of the asymptotes of curve C_1.
b) Describe the transformations that map curve C_1 to curve C_2.
c) Determine the equations of the asymptotes of curve C_2, giving your answer in the form $y = mx + c$, where m and c are constants to be found.
d) Find where the curve C_2 intersects with the coordinate axes.
e) Hence, sketch the curve C_2.

Question 15. A curve C has equation $\frac{x^2}{9} + \frac{y^2}{16} = 1$.

a) Sketch the curve C.

b) State two transformations that map C onto a circle of radius 1 with centre $(0, 0)$.

c) Find the area enclosed by the curve C.

The curve C is translated by vector $\begin{pmatrix} p \\ q \end{pmatrix}$, where p and q are positive constants, to form a new curve D. The curve D has a single point of contact with both the x-axis and the y-axis.

d) Find the values of p and q.

The curve C is stretched parallel to the x-axis by scale factor 2 to form curve E.

e) Find the equation of the transformed curve E in the form $ax^2 + by^2 = c$, where a, b and c are integers.

Question 16. The curve C_1, given by equation $\frac{x^2}{16} + y^2 = 1$, has two transformations applied to it followed by a translation to curve C_2 with equation $6x^2 + 24x + 2y + y^2 = -19$.

a) Find where the curve C_1 intersects with the coordinate axes.

b) Describe the transformations that map curve C_1 to curve C_2.

c) Find where the curve C_2 intersects with the coordinate axes.

d) By using transformations, find the values of the maximum and minimum values of x and y on C_2.

e) Hence, sketch the curve C_2.

Question 17. The curve C_1 has equation $\frac{x^2}{4} - \frac{y^2}{25} = 1$.

a) Sketch the curve C_1.

The curve C_1 is translated by vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$, where k is a positive constant, followed by a reflection in the line $y = -x$, to give curve C_2. The curve C_2 passes through the origin.

b) Determine the value of k.

c) Find the equations of the asymptotes of C_2, giving your answers in the form $y = mx + c$, where m and c are constants to be found.

Question 18. A hyperbola has equation $\frac{x^2}{4} - \frac{y^2}{3} = 1$. The hyperbola is transformed as follows: translated by vector $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$, followed by a rotation anti-clockwise by $\frac{\pi}{2}$, followed by a stretch in the x-direction by factor 3.

a) Show that the transformed hyperbola can be written in the form $4x^2 - 27y^2 + ax + by + c = 0$, where a, b and c are constants to be found.

b) Given that the line $y = k$ intersects the new hyperbola at two points, find the possible values of k, giving your answer in set notation.

Question 19. A hyperbola H has equation $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

a) Show that the line with equation $4x - 3y = 11$ intersects H at only one point.

The hyperbola H is translated by the vector $\begin{pmatrix} a \\ -1 \end{pmatrix}$ to form a new curve K. The line $4x - 3y = 11$ is now an asymptote to the translated curve K.

b) Find the value of a.

Question 20. The curve C_1 has equation $\frac{x^2}{25} + \frac{y^2}{4} = 1$. The curve C_2 is the image of C_1 after a reflection across the line $y = x$, followed by a translation by vector $\begin{pmatrix} a \\ b \end{pmatrix}$ for positive constants a and b, followed by an enlargement of scale factor 2 about $(0, 0)$. C_2 touches the x-axis at a single positive value of x and it touches the y-axis at a single positive value of y.

a) Find the value of a and b.

b) Find the equation of C_2 for your values of a and b, giving your answer in the form $\frac{(x-p)^2}{r} + \frac{(y-q)^2}{s} = 1$, where p, q, r and s are constants to be found.

Question 21. The curve C_1 has equation $\frac{(x-2)^2}{4} + \frac{y^2}{9} = 1$. The curve C_2 is the image of C_1 after a translation by vector $\begin{pmatrix} a \\ b \end{pmatrix}$ for non-zero negative constants a and b, followed by a reflection in the y-axis. Curve C_2 touches the curve C_1 at a single point and also touches an axis once.

a) State the name given to the curve C_1.
b) Find the values of a and b.
c) Find the equation of C_2.
d) State the point of intersection of the curves C_1 and C_2.
e) Sketch both C_1 and C_2 on the same axes.
f) Find the equations of the tangent lines that are a tangent line to both C_1 and C_2.

Question 22. An ellipse E has equation $\frac{x^2}{4} + \frac{y^2}{2} = 1$.

a) Write down the coordinates of the points where the ellipse E intersects the coordinate axes.

The ellipse E is translated by the vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$, where k is a positive constant, followed by an enlargement of scale factor $\sqrt{3}$ about the origin.

b) Find the equation of the transformed ellipse in the form $\frac{(x-a)^2}{p} + \frac{(y-b)^2}{q} = 1$, where a, b, p and q are constants to be found, giving your answer in terms of k.

A straight line L has equation $y = -\frac{1}{2}x + 5$.

c) Show that if the transformed ellipse intersects the line L then the points of intersection must satisfy $3x^2 - (4\sqrt{3} + 20)x + 6k^2 + 72 = 0$.
d) Given that the line L is a tangent to the translated ellipse, find the possible values of k.

Question 23. A hyperbola H has equation $\frac{x^2}{25} - \frac{y^2}{4} = 1$.

a) Sketch the hyperbola H.

It is known that the line with equation $y = x + k$ for constant k intersects the hyperbola H at two distinct points.

b) Show that $k > \sqrt{21}$ and $k < -\sqrt{21}$.

H is translated by the vector $\begin{pmatrix} a \\ b \end{pmatrix}$, where a and b are non-zero constants, to form another hyperbola whose equation is $4x^2 - 25y^2 - 32x + 50y = c$.

c) Find the values of the constants a, b and c.

d) Hence find an equation for each of the two tangents to $4x^2 - 25y^2 - 32x + 50y = c$ that are parallel to the line $y = x$.

Question 24. An ellipse E has equation $\frac{x^2}{4} + \frac{y^2}{100} = 1$. The line with equation $y = 2x + k$ for constant k intersects E at two distinct points.

a) Show that $-\sqrt{116} < k < \sqrt{116}$.

E is translated by vector $\begin{pmatrix} a \\ b \end{pmatrix}$, where a and b are non-zero constants, to form a curve with equation $25x^2 + y^2 - 150x + 10y = c$.

b) Find the values of the constants a, b and c.

c) Hence find an equation for each of the two tangents to $25x^2 + y^2 - 150x + 10y = c$ that are parallel to the line $y = 2x$.

Question 25. A curve C_1 has equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a) State the name given to the curve C_1.

b) Sketch the curve C_1.

The curve C_1 is transformed onto curve C_2 by a combination of a translation followed by a stretch. The equations of the asymptotes of C_2 are $y = 3x - k$ and $y = k - 3x$, for positive constant k.

c) In the case that $k = 6$, find the equation of C_2.

d) Determine the value of k such that C_1 and C_2 intersect at exactly three points.